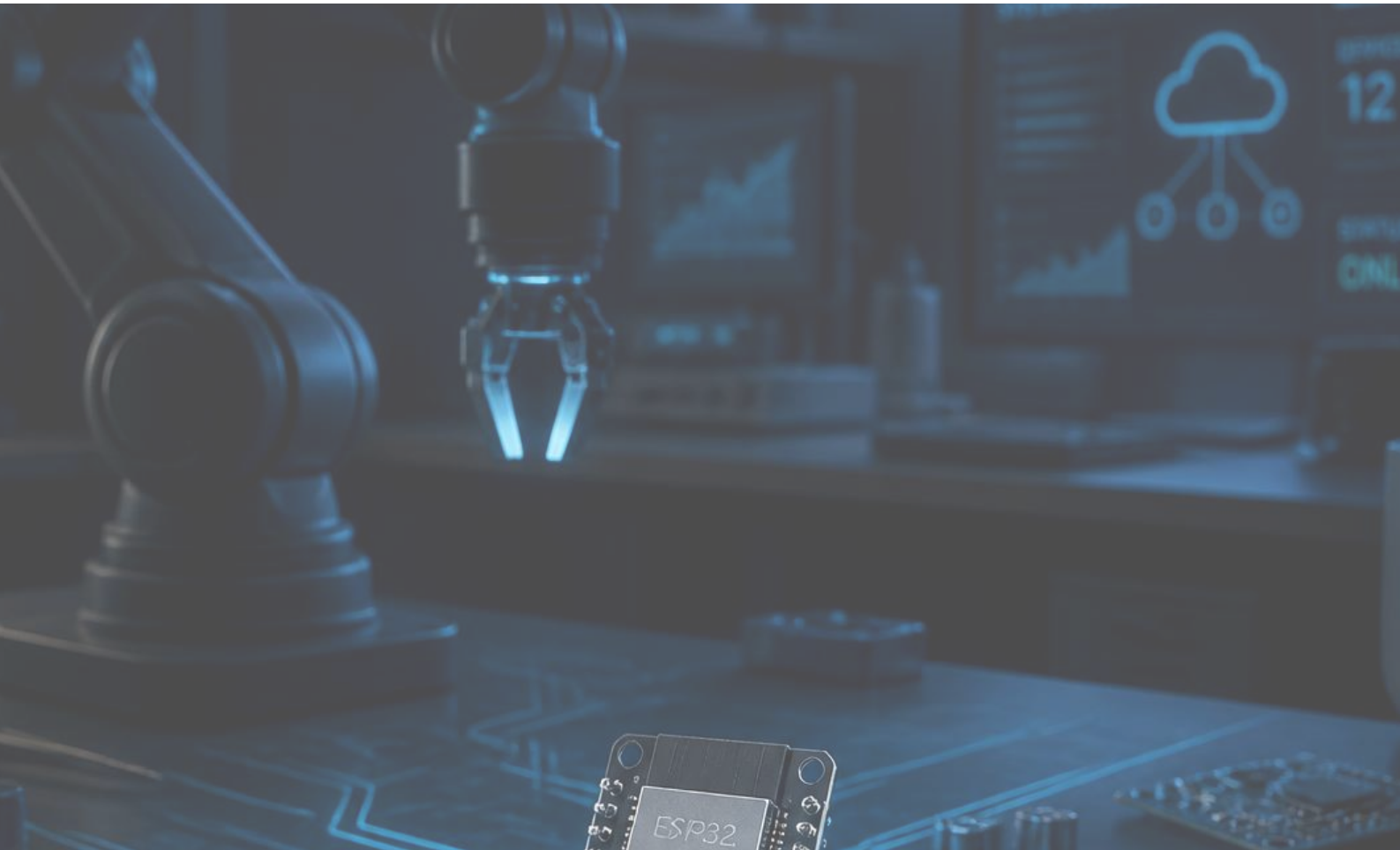




EDUFYI
TECH SOLUTIONS

JOB BRIDGE PROGRAMME 2026
16 WEEKS • 160 HOURS • 20+ PROJECTS



IoT, ROBOTICS & SMART MOBILITY

A 16-week, industry-oriented engineering programme covering embedded systems, edge AI, industrial IoT, autonomous robotics and electric-vehicle technology.

EDGE AI / TinyML

ROS 2 & SLAM

INDUSTRIAL IoT

EV & BMS

PROGRAMME AT A GLANCE

16 WEEKS

Structured, sprint-based delivery

160 HOURS

Live labs, studio time & reviews

20+ PROJECTS

Hardware builds and cloud stacks

1 CAPSTONE

Industry jury demo day

OUTCOMES



Embedded engineering

Bare-metal C, FreeRTOS tasks, low-power design and board bring-up.



Cloud & data pipelines

MQTT, REST, time-series storage, Grafana dashboards and alerting.



Edge intelligence

TinyML models trained, quantised and deployed on microcontrollers.



Autonomy stack

ROS 2 nodes, SLAM, sensor fusion and closed-loop motion control.



Smart mobility

EV powertrains, BMS logic, CAN networks and charging intelligence.



Secure by design

TLS, secure boot, signed OTA and device-identity management.

Built for the 2026 hiring market

Every module ends with a shippable artefact — schematic, repository, dashboard or demo video — so the portfolio is complete before placement season begins.

WHY EDUFYI

Transforming education into opportunity — with mentorship, certification and placement support engineered into the programme, not bolted on afterwards.



100% Placement Assistance

Interview drives, referrals and mock technical rounds until placed.



Global Certifications

Additional credentials from Microsoft, IBM and Cisco pathways.



Industry Mentors

Weekly reviews with practising embedded, robotics and EV engineers.



Lifetime Recordings

Every session recorded — revise labs anytime, anywhere.



24x7 Doubt Support

Dedicated engineering helpdesk and peer channels.



Resume & Aptitude Grooming

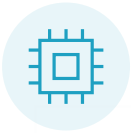
Portfolio reviews, GitHub polish and aptitude drills.

“TRANSFORMING EDUCATION INTO OPPORTUNITY”

EDUFYI TECH SOLUTIONS • BENGALURU

THE LEARNING PATH

MONTH 01



Electronics & Embedded Core

- Electronics, simulation & breadboarding bootcamp
- Sensor interfacing, ADC, filtering & calibration
- Wi-Fi, BLE and FreeRTOS multitasking
- Arduino / ESP32 architecture and Embedded C
- Motors, actuators, PWM and H-bridge drives

MONTH 02



Connectivity, Cloud & IIoT

- MQTT, Sparkplug B, Node-RED and Firebase
- Edge gateways, Docker, InfluxDB + Grafana
- Device security: TLS, secure boot, signed OTA
- LoRaWAN, NB-IoT, Matter/Thread & BLE Mesh
- Industrial IoT: OPC-UA, Modbus, digital twins

MONTH 03



Robotics, Motion & Edge AI

- ROS 2 nodes, topics, Gazebo simulation
- SLAM and Nav2 autonomous navigation
- TinyML + computer vision on edge hardware
- PID, encoders, odometry & IMU sensor fusion
- Robotic arms, kinematics and pick-and-place

MONTH 04



EV, Smart Mobility & Capstone

- EV / HEV architectures and powertrain sizing
- CAN bus, diagnostics and ADAS sensing basics
- Capstone integration, career prep & demo day
- Battery chemistry, BMS, SoC/SoH & thermal safety
- AI traffic control and smart EV charging grids

Weekly rhythm

2 live lab sessions · 1 mentor review · 1 build sprint · 1 graded deliverable — every single week, with recordings and code templates provided.

THE TECHNOLOGY STACK

Learners work on the exact toolchain used in production embedded, robotics and mobility teams.

HARDWARE

- ESP32 / ESP32-S3
- Arduino Uno & Nano
- Raspberry Pi 5
- STM32 Nucleo
- Jetson Nano (Edge AI)
- LoRa SX1278 · RFID · IMU

FIRMWARE

- Embedded C / C++
- FreeRTOS tasks & queues
- MicroPython
- PlatformIO
- OTA + secure boot
- Low-power & deep sleep

CONNECTIVITY

- MQTT + Sparkplug B
- LoRaWAN / TTN
- BLE Mesh · Matter · Thread
- Modbus TCP · OPC-UA
- CoAP & WebSockets
- NB-IoT overview

CLOUD & DATA

- Node-RED flows
- Firebase Realtime DB
- AWS IoT Core basics
- InfluxDB time series
- Grafana dashboards
- REST & webhook design

ROBOTICS & AI

- ROS 2 Humble
- Gazebo & RViz
- SLAM + Nav2
- OpenCV & YOLO edge models
- TensorFlow Lite Micro
- Edge Impulse TinyML

MOBILITY & EDA

- CAN bus & diagnostics
- BMS SoC / SoH logic
- MATLAB-style modelling
- KiCad PCB design
- Thermal simulation
- Digital twin dashboards

Lab kit & environment

Each learner receives a hardware kit and cloud workspace so every lab can be reproduced at home.

- ESP32 dev board + sensor pack
- OLED, relay & actuator set
- Cloud broker + Grafana workspace
- Motor driver, servos & chassis
- Breadboard, jumpers & multimeter
- Simulator licences (Wokwi/Gazebo)

MODULE 01

WEEKS 0 - 4

Electronics, Microcontrollers & Sensors

WEEK 00

Foundation Bootcamp

- IoT & robotics landscape, career map
- Circuit safety, breadboarding, DMM use
- Tinkercad and Wokwi simulation
- Ohm's law, dividers, pull-ups

PROJECT First Blink Challenge

WEEK 01

Microcontrollers & Embedded C

- Board architecture, GPIO, registers
- Debouncing and state machines
- Digital I/O, timers, interrupts
- Toolchain: PlatformIO + Git basics

PROJECT Digital Doorbell System

WEEK 02

Sensors & Signal Conditioning

- Analog vs digital, ADC resolution
- Noise filtering and moving averages
- Ultrasonic, IR, DHT11, LDR, MQ-series
- I2C / SPI bus interfacing

PROJECT Smart Room-Light & Air Quality Node

WEEK 03

Motors, Drives & Power

- Servo PWM and torque calculation
- Battery budgeting & regulator design
- L298N H-bridge and stepper drives
- Multi-servo arm simulation

PROJECT Line-Follower / Obstacle-Avoider Bot

WEEK 04

Wireless & RTOS Foundations

- ESP32 Wi-Fi station & AP modes
- BLE and Bluetooth Classic app control
- FreeRTOS tasks, queues, dual-core
- Firebase live data streaming

PROJECT Phone-Controlled Robot + Live Dashboard

MODULE 02

WEEKS 5 - 8

Wireless IoT, Cloud, IIoT & Security

WEEK 05

IoT Automation & Visualisation

- Node-RED flows, widgets, gauges
- InfluxDB time-series + Grafana
- Webhooks and rule-based triggers
- Alerting via email / Telegram bots

PROJECT IoT Temperature-Alert & Analytics System

WEEK 06

Protocols & Wireless Networks

- IPv6, 6LoWPAN and addressing
- LoRaWAN class A/B/C with TTN
- BLE Mesh, Zigbee, Matter & Thread
- MQTT broker, QoS, retained topics

PROJECT Multi-Node LoRa Sensor Network

WEEK 07

Edge Infrastructure & Security

- WebSocket servers on ESP8266
- TLS, X.509 identity, secure boot
- Raspberry Pi edge gateway with Docker
- Signed OTA firmware updates

PROJECT Secure Edge Gateway with OTA

WEEK 08

Industrial IoT & Digital Twins

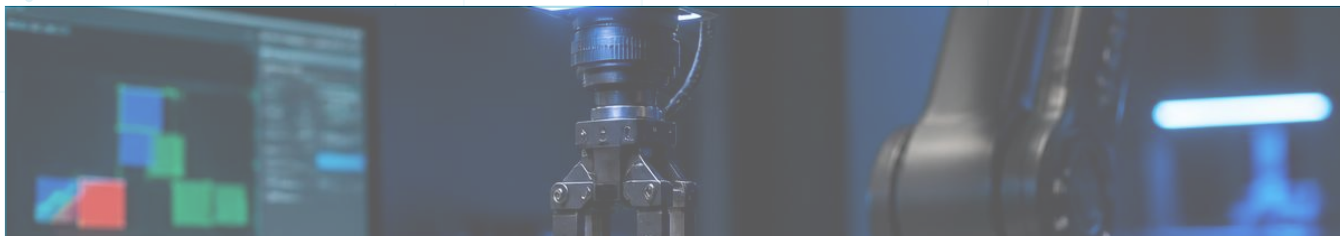
- Smart factory & ISA-95 layers
- MQTT Sparkplug B payloads
- OPC-UA and Modbus TCP integration
- Predictive maintenance with ML anomalies

PROJECT Digital-Twin Machine Health Dashboard

MODULE 03

WEEKS 9 - 12

Robotics, Motion Control & Embedded AI



WEEK 09

ROS 2 & Robot Architecture

- ROS 2 nodes, topics, services
- Gazebo simulation workflows
- URDF modelling and RViz
- Voice & gesture command interfaces

PROJECT Gesture-Controlled Robotic Car

WEEK 10

Autonomous Navigation

- LiDAR scanning and mapping
- Obstacle costmaps & recovery behaviour
- SLAM with Nav2 path planning
- Warehouse fleet logic

PROJECT Autonomous Warehouse Navigation Bot

WEEK 11

Motion Control & Sensor Fusion

- Open vs closed-loop control
- IMU + Kalman / complementary filters
- Encoders, odometry, dead reckoning
- PID tuning with Serial Plotter

PROJECT Precision PID Race Bot

WEEK 12

Embedded AI & Computer Vision

- Forward / inverse kinematics of arms
- YOLO object detection at the edge
- OpenCV detection on Raspberry Pi
- TinyML with TF Lite Micro & Edge Impulse

PROJECT Vision-Guided Object Sorting Arm

MODULE 04

WEEKS 13 - 16

EV, Smart Mobility, Capstone & Careers



WEEK 13

Electric Vehicle Fundamentals

- EV / HEV / PHEV architectures
- Super-capacitors, C-rate, range math
- Li-ion, LFP, solid-state & metal-air
- Powertrain sizing and efficiency

PROJECT EV vs IC Powertrain Analysis Report

WEEK 14

Battery Management & Motors

- BMS: cell balancing, SoC / SoH estimation
- PMSM, BLDC, axial-flux motor types
- Thermal runaway, active & passive cooling
- CAN bus messaging & diagnostics

PROJECT BMS Telemetry & Thermal Safety Design

WEEK 15

Smart Mobility & Connected Infrastructure

- Vehicle density detection with vision
- V2X and telematics fundamentals
- Adaptive AI traffic-signal control
- Smart charging & grid load prediction

PROJECT AI-Balanced EV Charging Station

WEEK 16

Capstone & Career Launch

- System architecture and scoping
- Technical documentation & GitHub polish
- Hardware + firmware + cloud integration
- Mock interviews and jury presentation

PROJECT Capstone Demo Day — Industry Jury Panel

ADVANCED ELECTIVE TRACKS

Beyond the core sixteen weeks, learners choose two advanced tracks to deepen expertise and differentiate their capstone portfolio.



Edge AI & TinyML

Model quantisation, keyword spotting, vibration classification and on-device inference with TF Lite Micro.



ROS 2 Autonomy

Multi-node architectures, SLAM, Nav2 planners and Gazebo digital-twin simulation of mobile robots.



IoT Cybersecurity

Threat modelling, device identity, TLS mutual auth, secure boot, flash encryption and signed OTA.



Cloud & Data Engineering

AWS IoT Core device shadows, MQTT bridges, time-series pipelines and Grafana observability.



LPWAN & Private Networks

LoRaWAN gateway deployment, NB-IoT, Matter/Thread border routers and RF link budgeting.



EV Systems & CAN

BMS algorithms, CAN/CAN-FD diagnostics, charging protocols and V2X connected-vehicle basics.



PCB & Product Design

KiCad schematic capture, layout, DFM rules, enclosure fit and certification readiness.



Medical & Wearable IoT

Biosignal acquisition, ECG/SpO2 pipelines, BLE profiles and data-privacy considerations.

How electives run

Tracks run as evening deep-dive labs in weeks 9-16, each with its own mini-project that plugs directly into the capstone architecture. Track certificates are issued separately.

20+ INTEGRATED PROJECTS

Six project families, each ending in a documented, demonstrable build.



Electronics & Embedded

- Smart doorbell & access node
- Multi-sensor data logger
- Low-power battery node



Cloud & Data Visualisation

- Live MQTT dashboard
- Grafana analytics stack
- Anomaly alert engine



Medical IoT & Healthcare

- Patient vitals monitor
- Wearable fall detector
- Cold-chain vaccine tracker



Robotics & Autonomy

- Vision sorting arm
- SLAM warehouse bot
- PID racing platform



EV & Smart Mobility

- BMS telemetry unit
- AI charging balancer
- Adaptive traffic controller



Smart Agriculture & City

- Precision irrigation node
- Air-quality mesh
- Smart street lighting

EVERY PROJECT SHIPS WITH

Circuit schematic

Documented repo

Live dashboard

Demo video

Capstone: an end-to-end connected system

Sensors → edge inference → secure transport → cloud analytics → actuation, presented to an industry jury panel.

ROLES YOU CAN TARGET

Embedded Systems Engineer

Robotics Engineer

Automation & Control Engineer

Computer Vision Engineer

Smart Mobility Engineer

Medical IoT Engineer

FPGA & Digital Systems Engineer

BMS & Battery Systems Engineer

IoT Solutions Engineer

Industrial IoT (IIoT) Engineer

Edge AI / TinyML Engineer

EV Systems Engineer

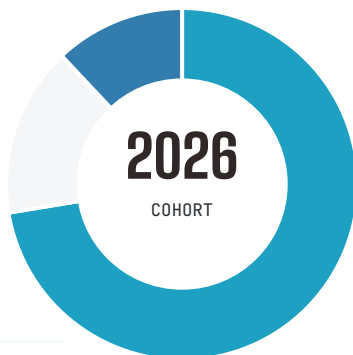
LoRaWAN / Wireless Engineer

PCB & Product Development Engineer

IoT Cloud & Data Analytics Engineer

Research & Product Development Engineer

LEARNER BACKGROUND



- 72.4%** UG Management
- 15.6%** UG B.Tech / BE
- 12.0%** Others

Placement engine

Resume and GitHub review · aptitude and DSA drills · mock technical interviews · hiring-partner drives · post-programme mentoring until placed.

ACCREDITED CERTIFICATIONS.

Every learner receives industry-recognised, accredited certificates on completion of the internship, industrial training and capstone project modules.



INTERNSHIP CERTIFICATE
Accredited Certification



INDUSTRIAL TRAINING
Accredited Certification



PROJECT COMPLETION
Accredited Certification

● INTERNSHIP

Verified internship completion credential

● TRAINING

Industrial training hours certified

● PROJECT

Capstone project completion proof

GLOBAL MNC CERTIFICATIONS.

Learners are prepared for and certified through globally recognised professional certification programs from leading technology companies.



CISCO
Cybersecurity CCST



MICROSOFT
Azure Fundamentals



IBM
Internet of Things



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ESB
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